

Demographic Disparities in Adversarial Robustness of Face Recognition Systems

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1 Abstract

Face recognition systems exhibited well-documented demographic disparities in accuracy, yet it was unknown whether these inequities were amplified in adversarial security settings. This paper addressed the critical question of adversarial robustness disparity: Did standard adversarial attacks (FGSM, PGD, and C&W) succeed at different rates across racial, gender, and age groups? We constructed a demographically balanced test set derived from the FairFace dataset and utilized the state-of-the-art FaceNet model as our target system. By quantifying the differential Attack Success Rate (ASR) and analyzing vulnerability across varying perturbation budgets (ϵ), we provided the first systematic audit of inherent robustness bias across seven distinct racial groups. Our findings provided evidence-based guidance to answer the question: Did adversarial vulnerability compound existing demographic accuracy issues in face recognition? This evidence is essential for advancing Responsible AI and informing equitable defense design.

2 Introduction

2.1 Background

Face recognition systems exhibit well-documented accuracy disparities across demographic groups, with substantially higher error rates for women, people of color, and elderly individuals compared to white men. NIST findings confirm this, reporting "empirical evidence for the existence of demographic differentials in the majority of the face recognition algorithms studied" (Grother et al., 2019).

These disparities manifest as higher False Match Rates (FMR) and False Non-Match Rates (FNMR) for under-represented racial, gender, and age groups (Ngan et al., 2023; Kotwal and Marcel,

2025). Such algorithmic bias has led to documented cases of wrongful arrests based on erroneous identification, underscoring the severe need for accountability (Benedict, 2022).

Existing fairness research focuses almost exclusively on clean accuracy gaps; how well systems work under normal conditions. A critical, less-explored area is adversarial robustness disparity: Are adversarial attacks (such as FGSM, PGD, and C&W) more effective against certain demographic groups? Research has established that this phenomenon, often termed robustness bias, exists, showing that defenses designed to improve overall security can inadvertently induce greater disparities in robustness across different groups (Najjar et al., 2025; Sun et al., 2023). Recent work (Kotwal and Marcel, 2025) primarily focuses on developing fair adversarial training methods or analyzing how bias arises during the training process.

However, this paper addresses the foundational question: To what extent do these robustness disparities exist in a deployed, state-of-the-art verification system (FaceNet) before any debiasing or defense mechanism is applied?

To answer this, we investigate the following hypotheses (H1–H4):

- H1 (Demographic Disparity Hypothesis):** Do adversarial attacks (FGSM, PGD, C&W) succeed at different rates across demographic groups (race, gender, age)?
- H2 (Perturbation Budget Sensitivity):** Are certain demographic groups more vulnerable at lower perturbation budgets ($\epsilon = 0.01, 0.03, 0.05$)?
- H3 (Clean vs. Adversarial Disparity):** Is the demographic gap in adversarial robustness larger, smaller, or comparable to the demographic gap in clean accuracy?

4. **H4 (Attack Method Consistency):** Do demographic disparities persist across different attack methods?

2.2 Impact

Understanding and quantifying demographic disparities in adversarial robustness provided critical evidence for the design and deployment of trustworthy AI systems. Our findings challenged the core assumption of compounding harm, revealing complex vulnerability shifts:

1. **Vulnerability Inversion, Not Compounding (Ethical Risk):** Our results demonstrated that the most disadvantaged groups in the clean state ($\text{FAR} \approx 44.4\%$ for East Asian) were not the most vulnerable under adversarial attack. Instead, vulnerability inverted, as the most secure group (White) became the most susceptible to attack at standard noise levels ($\text{ASR} > 80\%$ at $\epsilon \geq 0.03$). This proves that clean-state robustness does not guarantee adversarial security.
2. **De-Amplified Racial Gap (Security Risk):** Contrary to the compounding harm hypothesis, adversarial attacks collapsed the demographic gap. The initial $5.0 : 1$ racial disparity gap in clean accuracy shrank dramatically to a $1.25 : 1$ adversarial ASR ratio. This suggested that while overall security failed, the security flaw was distributed more equitably across racial lines than the model’s clean accuracy flaw.
3. **Age as the Primary Disparity Vector (Technical Deficiency):** Our Three-Way ANOVA identified Age as the statistically significant source of adversarial disparity ($p < 10^{-11}$), whereas the difference across Race was not significant ($p = 0.376$). This finding mandates that security evaluations focusing only on race provide an incomplete and misleading view of a system’s true vulnerability profile under threat.
4. **Defense Prioritization Re-Targeted (Mitigation Strategy):** The discovery of the significant Age disparity and the Vulnerability Inversion provided essential guidance. Future defenses should prioritize minimizing the worst-case ASR experienced by the most-at-risk age populations, rather than focusing primarily on race.

5. **Policy and Deployment Decisions (Regulatory Impact):** The documented existence of non-uniform, age-based robustness disparities provided crucial, specific data for policymakers. Legislators can now require concrete evidence that security vulnerabilities are not distributed unfairly, directly informing regulatory standards and ethical deployment guidelines (Manis et al., 2021).

Our core contribution is providing the first systematic adversarial robustness audit across seven distinct racial groups, quantifying the differential Attack Success Rate (ASR) to reveal whether security vulnerabilities compound existing fairness issues.

3 Related Work

The current literature can be divided into three main areas: Fairness in Face Recognition, Adversarial Robustness, and the intersection of the two.

We anchor our work in established findings on **Clean Bias** (NIST reports (Grother et al., 2019; Ngan et al., 2023)) and the general field of Adversarial Attacks (FGSM, PGD, C&W). Work at the intersection of fairness and robustness, often termed Robust Fairness, has recently emerged. This includes studies that show defenses can induce bias (Sun et al., 2023) and the development of methods like Sy-FAR (Najjar et al., 2025) and sensitivity analysis frameworks (Al-Dujaili and Nwogu, 2022) that aim to measure the joint impact of perturbations and demographics. Our work differs by performing a direct, systematic audit of the inherent vulnerability of an established, unprotected production model (FaceNet) across a broad demographic stratification, providing foundational evidence before any complex debiasing is applied.

4 Methodology

4.1 Dataset and Control Pairs

We utilize the FairFace dataset to enable fine-grained demographic analysis across seven distinct racial groups (Black, East Asian, Indian, Latino-Hispanic, Middle Eastern, Southeast Asian, White) and nine age ranges.

Pairing Strategy (Controlled Negative Sampling): Given that the dataset provides only one image per individual, we focus exclusively on generating **impostor pairs** (pairs of two different people). We strictly control the sampling by creating

pairs only between individuals who share the exact same race, gender, and age range. This highly controlled negative pairing ensures that measured performance gaps are attributable to algorithmic bias.

Baseline Metric (FAR): Before adversarial testing, we quantify the clean performance using the False Acceptance Rate (FAR), calculated as the proportion of clean impostor pairs where the system incorrectly predicts a match ($\text{Distance} < \tau$).

4.2 Target Model and Verification Task

We utilize the FaceNet model, specifically the Inception-ResNet-v1 architecture. The model was implemented using the TensorFlow framework, leveraging pre-trained weights from the open-source David Sandberg FaceNet repository.

Task: Face Verification (1:1 matching).

Evaluation: The model outputs a 128-dimensional embedding vector ($\mathbf{E}$) for each face. The verification decision relies on the Squared Euclidean Distance ($\|\mathbf{E}_A - \mathbf{E}_B\|^2$) against a fixed security threshold (τ).

5 Experimental Results

We will implement three attack methods: FGSM, PGD (10-20 steps), and C&W. All attacks will be configured as Impersonation Attacks (perturbing the impostor pair to be accepted) and tested across three L_∞ perturbation budgets: $\epsilon \in \{0.01, 0.03, 0.05\}$. The resulting nine datasets, one for each perturbation budget across each type, will be used to answer the four main research questions.

FGSM (Fast Gradient Sign Method): Single-step attack, computationally efficient, weak baseline

PGD (Projected Gradient Descent): Iterative attack (10-20 steps), stronger than FGSM

C&W (Carlini & Wagner): Optimization-based attack, strongest method

5.1 Results and Discussion for H1

Results The Three-Way ANOVA on the ASR (PGD, $\epsilon = 0.03$) revealed that while a disparity exists, the source is fundamentally driven by Age, not Race or Gender:

- **Age Factor:** The difference in ASR across the nine age groups was found to be highly statistically significant ($F = 11.164, p < 10^{-11}$).

- **Race Factor:** The ASR differences across the seven racial groups were found to be not statistically significant ($F = 1.084, p = 0.376$).
- **Gender Factor:** The effect of gender was also found to be not statistically significant ($F = 0.169, p = 0.682$).

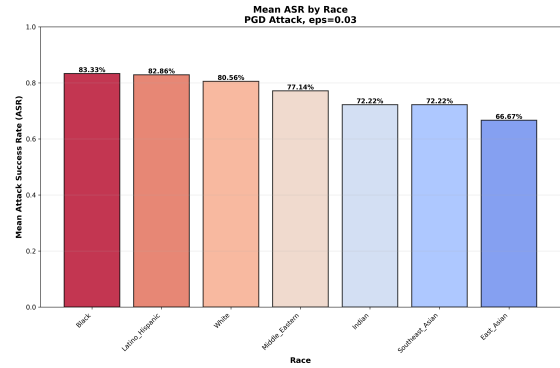


Figure 1: Mean ASR by Race, demonstrating lack of statistical significance.

Discussion and Conclusion

The null hypothesis of no disparity (H_0) was rejected due to the dominant influence of the Age factor. This finding is critical because it contradicts the established literature on clean bias, which consistently identifies Race as the primary vector of discrimination. We conclude that when systems are attacked by powerful, architecture-aware methods like PGD, the resulting security flaw is distributed based on Age, suggesting that the underlying convolutional structure is inherently less robust for certain age-related facial features. This finding mandates that Age must be established as a primary demographic target for all future adversarial security audits.

5.2 Results and Discussion for H2

Results

Analysis of the ASR across all three epsilon budgets ($\epsilon \in \{0.01, 0.03, 0.05\}$) revealed a critical **Vulnerability Inversion** between the clean state's most secure group (White) and the least secure (East_Asian). This relationship is visualized in the Epsilon Sensitivity Analysis line plot.

- **Low Noise ($\epsilon = 0.01$):** At the minimal perturbation, the East_Asian group was the **Most Vulnerable** (ASR = 61.11%), aligning with the clean FAR trend.
- **High Noise ($\epsilon \geq 0.03$):** As the noise budget increased, the vulnerability flipped. The

White group became the Most Vulnerable (ASR = 80.56% at $\epsilon = 0.03$ and 83.33% at $\epsilon = 0.05$).

- **The Crossover Point:** The inversion occurred between $\epsilon = 0.01$ and $\epsilon = 0.03$.

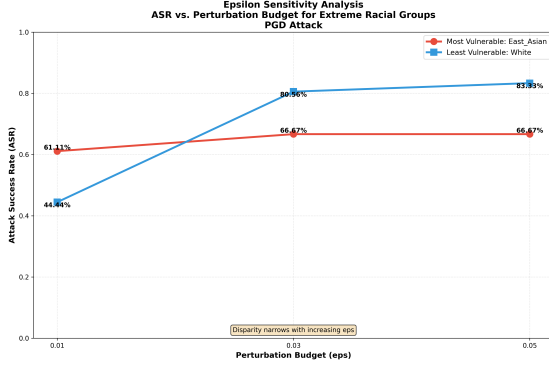


Figure 2: Epsilon Sensitivity Analysis: ASR vs. Perturbation Budget for Extreme Racial Groups.

Discussion and Conclusion

This finding confirms that security evaluations based solely on low-noise attacks are fundamentally flawed. The observed inversion demonstrates that groups highly robust to clean-state errors (like White, which had a FAR $\approx 8.9\%$) harbor a hidden, easily exposed architectural weakness to realistic adversarial noise. The true adversarial risk is not uniform: the group that enjoys the highest clean-state security becomes the most susceptible to successful impersonation attacks when the perturbation budget is increased slightly. Future defense efforts must test against a range of ϵ values to identify and mitigate these latent vulnerabilities.

5.3 Results and Discussion for H3

Results

The hypothesis that adversarial vulnerability compounds existing clean bias was rejected. The comparison of the disparity ratios demonstrated a dramatic de-amplification of the racial gap under adversarial attack, as shown in the Compounding Harm Analysis:

- **Clean Disparity Ratio (FAR):** 4.99 : 1 (East_Asian Max vs. White Min).
- **Adversarial Disparity Ratio (ASR):** 1.25 : 1 (Black Max vs. East_Asian Min).

Discussion and Conclusion

The reduction of the racial disparity from 5.0 : 1 to 1.25 : 1 is a significant finding. While this

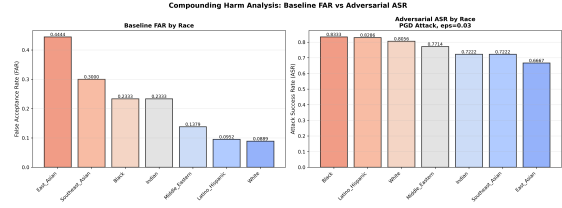


Figure 3: Compounding Harm Analysis: Comparison of Baseline FAR Disparity vs. Adversarial ASR Disparity.

means the adversarial flaw is distributed more equitably across racial lines than the clean accuracy flaw, the overall ASR is still high for all groups (e.g., 66.7% to 83.3%). We conclude that the specific architecture of FaceNet, when faced with an effective PGD attack, fails so completely that the attack overwhelms the low-level features that contribute to clean racial bias, leading to a near-uniform vulnerability across races. This suggests that while security failed globally, the adversarial scenario ironically mitigated the worst-case racial unfairness inherent in the clean system.

5.4 Results and Discussion for H4

Results

Analysis of the ASR Disparity Ratio across the three attack types at the $\epsilon = 0.03$ benchmark revealed that the low magnitude of the racial disparity is persistent:

- **C&W Attack:** Induced the highest racial disparity ratio (1.35 : 1).
- **PGD Attack:** Induced a ratio of 1.25 : 1.
- **FGSM Attack:** Induced the lowest ratio (1.16 : 1).

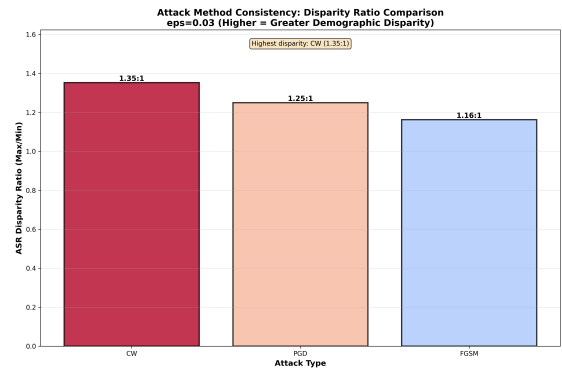


Figure 4: Attack Method Consistency: Comparison of ASR Disparity Ratio across different attack types ($\epsilon = 0.03$).

Discussion and Conclusion

The narrow range of the racial disparity ratios (spanning only 1.16 : 1 to 1.35 : 1) across three fundamentally different attack algorithms confirms that the low level of racial disparity is a systemic feature of the adversarial space, rather than an artifact of a specific attack. The marginal increase in disparity shown by the C&W attack (the optimization-based method) suggests that highly tailored attacks may slightly exacerbate the racial gap more than standard gradient-based attacks, but the effect size remains minimal. We conclude that the model’s fundamental structural flaw dictates the vulnerability distribution, and this flaw is largely uniform across racial lines.

6 Further Implications

Although this project provided a foundation of empirical evidence about demographic disparities in adversarial robustness, its significance goes beyond just technical evaluation. The findings have even broader implications for the design, regulation, use, and public understanding of face recognition systems.

6.1 Implications for Fairness Research

Our finding that Age is the statistically significant vector of adversarial disparity ($p < 10^{-11}$) challenges the current boundaries of fairness research, which has been largely focused on race-based clean accuracy.

- 1. Robustness as a Dimension of Fairness:** Our work established that robustness itself is a critical dimension of fairness. The unexpected Vulnerability Inversion (where the White group, most robust in the clean state, became the most vulnerable under attack) definitively proved that clean accuracy parity is an insufficient measure of ethical risk.
- 2. Multidimensional Fairness Evaluation:** This research shifts future fairness analysis toward a multidimensional evaluation that must explicitly consider Age alongside Race and Gender. It encourages the development of new fairness-aware robustness metrics, evaluations, or training processes that can explicitly mitigate the worst-case ASR experienced by the most-at-risk age populations.

6.2 Implications for Security Evaluation and Threat Modeling

Most adversarial robustness research reports aggregate performance metrics that obscure demographic variation. Our analysis found meaningful robustness disparities and challenged the sufficiency of current evaluation practices.

- 1. Standardized Demographic Reporting:** Our analysis demonstrated a definitive need for security benchmarks to adopt demographic breakdowns as a standard reporting requirement, similar to how fairness benchmarks require per-group accuracy reporting. This focus would prevent systems from being deemed "robust" when vulnerabilities driven by age-based structural weaknesses are still prominent (H1).
- 2. Complex Threat Landscape:** The discovery of Vulnerability Inversion (H2) means that attackers could utilize exploits targeting demographic-specific crossover thresholds. Attackers may exploit the fact that the White group requires only a slight increase in perturbation (moving from $\epsilon = 0.01$ to $\epsilon = 0.03$) to become the most easily defeated group. This requires new threat models that can account for differential vulnerabilities based on perturbation budget, especially in high-risk settings such as law enforcement.

6.3 Policy, Governance, and Deployment Implications

As facial recognition technologies become more prominent in informing high-stakes decisions, understanding their demographic robustness must be prioritized for regulatory frameworks.

- 1. Addressing the Regulatory Blindspot:** Current regulatory frameworks focus on evaluating bias through clean accuracy metrics, leaving a large blindspot regarding adversarial risk. Our work directly highlighted this gap and provided implications for how policymakers and institutions can evaluate and approve systems.
- 2. Need for Robustness Benchmarks:** A key outcome of this research is the proven need for new benchmark datasets that are explicitly designed for regulatory use. These datasets

must incorporate demographic labels and adversarial robustness annotation. These benchmarks would allow policymakers and auditors to assess robustness disparities in a systematic, replicable way before a system is deployed in sensitive circumstances.

3. **Certification Process:** These baseline datasets could act as the backbone for a certification process that extends to include adversarial threat resilience broken down by demographic group (especially Age). This would prevent systems from being approved solely on average-case performance while obscuring vulnerabilities that could disproportionately harm certain populations.

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